

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Main Components Inside a Desktop PC

Component	Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	The brain of the computer that performs calculations and executes instructions.
RAM (Random Access Memory)	Temporary memory used to store data and programs currently in use.
Storage Drive (HDD/SSD)	Stores the operating system, software, and user data permanently.
Power Supply Unit (PSU)	Converts AC power from the wall outlet into DC power for computer components.
Graphics Card (GPU)	Processes graphics and video output for gaming, videos, and design applications.
CPU Cooler (Fan & Heatsink)	Removes heat from the processor to prevent overheating.
Case Fans	Improve airflow inside the cabinet and help keep components cool.
Optical Drive (DVD/CD Drive)	Reads and writes data on CDs and DVDs (available in some PCs).
Network Card (NIC)	Enables the computer to connect to a network or the internet.
Sound Card	Processes audio input and output for speakers, headphones, and microphones.
CMOS Battery	Maintains BIOS settings and system clock when the computer is powered off.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Step	Caption / Explanation
Step 1: Shut Down and Unplug Power	Turn off the computer completely and unplug the power cable from the wall socket. This prevents electric shock and protects components from damage. <i>(Insert Photo/Diagram 1)</i>
Step 2: Remove Side Panel	Unscrew the side panel screws and carefully slide the panel off to access the internal components. <i>(Insert Photo/Diagram 2)</i>
Step 3: Disconnect Internal Cables	Disconnect power cables, SATA cables, and front-panel connectors from the motherboard and storage devices. This makes component removal easier. <i>(Insert Photo/Diagram 3)</i>
Step 4: Remove RAM Modules	Press the locking clips on both ends of the RAM slots and gently lift out the RAM sticks. <i>(Insert Photo/Diagram 4)</i>
Step 5: Remove HDD/SSD	Unscrew or unclip the HDD/SSD from its mounting bay and disconnect any remaining cables. <i>(Insert Photo/Diagram 5)</i>
Step 6: Remove Graphics Card (if installed)	Unscrew the graphics card bracket, release the PCIe slot lock, and carefully pull out the card. <i>(Insert Photo/Diagram 6)</i>
Step 7: Remove Power Supply (PSU)	Unscrew the PSU from the case and carefully remove it after disconnecting all cables. <i>(Insert Photo/Diagram 7)</i>
Step 8: Remove Motherboard	Unscrew all motherboard mounting screws and carefully lift the motherboard out of the case. <i>(Insert Photo/Diagram 8)</i>
Step 9: Organize Components	Place all removed components and screws in separate containers to avoid loss or damage. <i>(Insert Photo/Diagram 9)</i>

3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step	Explanation
Step 1: Install the Motherboard	Place the motherboard inside the case and secure it with screws. The motherboard must be installed first because all other components connect to it. <i>(Insert Photo/Diagram 1)</i>
Step 2: Install the Power Supply (PSU)	Mount the PSU in its designated location and secure it with screws. The PSU provides power to all components. <i>(Insert Photo/Diagram 2)</i>
Step 3: Connect Motherboard Power Cables	Connect the 24-pin ATX power connector and CPU power connector to the motherboard. These connections are necessary before powering the system. <i>(Insert Photo/Diagram 3)</i>
Step 4: Install the CPU and CPU Cooler	Carefully place the CPU into the socket and attach the CPU cooler. The CPU must be installed before RAM and other components around it. <i>(Insert Photo/Diagram 4)</i>
Step 5: Install RAM Modules	Insert RAM sticks into the appropriate DIMM slots until the locking clips click into place. RAM is required for system operation. <i>(Insert Photo/Diagram 5)</i>
Step 6: Install HDD/SSD	Secure the storage drive in its bay and connect SATA power and data cables. Storage devices are needed to store the operating system and files. <i>(Insert Photo/Diagram 6)</i>
Step 7: Install Graphics Card (if available)	Insert the graphics card into the PCIe slot and secure it with screws. Connect PCIe power cables if required. <i>(Insert Photo/Diagram 7)</i>
Step 8: Connect Front Panel and Other Cables	Connect power switch, reset switch, USB, audio, and fan cables to the motherboard headers. These connections allow the PC case controls and ports to function. <i>(Insert Photo/Diagram 8)</i>
Step 9: Check All Connections	Verify that all cables and components are properly installed and secured. This helps prevent startup issues. <i>(Insert Photo/Diagram 9)</i>
Step 10: Close the Side Panel and Test	Reattach the side panel, connect external cables, and power on the system to ensure everything works correctly. <i>(Insert Photo/Diagram 10)</i>

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Checklist Item	Type	Why It Is Important
Turn off the computer	Safety Precaution	Prevents electrical shock and protects components from damage.
Unplug the power cable	Safety Precaution	Ensures that no electrical current is flowing through the system.
Wear an anti-static wrist strap	Safety Precaution	Prevents electrostatic discharge (ESD) that can damage electronic components.
Work on a clean and dry surface	Safety Precaution	Reduces the risk of dust, moisture, and accidental damage.
Keep liquids away from the work area	Safety Precaution	Prevents short circuits and hardware failure.
Handle components by their edges	Safety Precaution	Protects sensitive circuits and connectors from damage.
Organize screws and small parts	Safety Precaution	Prevents loss of important components during assembly/disassembly.
Phillips head screwdriver	Tool	Used for removing and tightening screws in the PC cabinet and components.
Anti-static wrist strap	Tool	Protects hardware from static electricity.
Flashlight	Tool	Helps clearly identify components inside the cabinet.
Small container for screws	Tool	Keeps screws and small parts organized and secure.
Cable ties	Tool	Helps manage cables neatly and improves airflow.
Soft brush / Compressed air can	Tool	Used to clean dust from components safely.
Thermal paste (if required)	Tool	Ensures proper heat transfer between the CPU and heatsink.